Orchestrate processes and data movement with Microsoft Fabric

Core Definition and Architecture

- **Definition**: Data pipelines are a sequence of connected activities that automate the overall process of extracting, transforming, and loading (ETL) data from various sources into analytical destinations.
- **Architecture**: Fabric pipelines use the same architecture as Azure Data Factory, utilizing connected activities and control flow logic to manage complex data tasks.
- **Visual Interface**: A graphical pipeline canvas allows users to build these workflows with minimal or no coding.

Primary Components

- **Activities**: These are executable tasks within a pipeline. They fall into two categories:
    - **Data Transformation**: Includes "Copy Data" (direct transfer), "Data Flow" (Dataflows Gen2 for complex logic), and "Notebook" (Spark code).
    - **Control Flow**: Manages pipeline logic such as loops, conditional branching, and managing variables or parameters.
- **Parameters**: Users can parameterize pipelines to provide specific values (e.g., folder names) during execution, which increases reusability and flexibility.
- **OneLake Integration**: Pipelines can load data into various Fabric stores, including lakehouses, warehouses, and SQL databases.

Key Activities and Tools

- **Copy Data Activity**: This is the most common pipeline use case for ingesting raw data from external sources.
- **Copy Data Tool**: A guided graphical assistant that walks users through selecting a connector for the source and defining the destination.
- **Pipeline Templates**: Microsoft Fabric provides predefined templates for common scenarios (e.g., bulk copy from a database, moving data from ADLS Gen2 to a Lakehouse) to speed up development.

Execution and Management

- **Validation and Runs**: Before execution, users can use the "Validate" option to check configurations. A "run" is initiated each time the pipeline executes.
- **Scheduling**: Pipelines can be run on-demand or scheduled to start automatically at specific frequencies.
- **Monitoring**: Users can view the run history to confirm success, investigate specific settings, or troubleshoot failures using unique run IDs.

Data Pipelines in Microsoft Fabric Assessment

1. What is a data pipeline?
A) A special folder in OneLake storage where data can be exported from a lakehouse.
B) A sequence of activities to orchestrate a data ingestion or transformation process.
C) A saved Power Query.
D) A physical hardware cable used to connect local servers to the Azure cloud.

2. You want to use a pipeline to copy data to a folder with a specified name for each run. What should you do?
A) Create multiple pipelines—one for each folder name.
B) Use a Dataflow (Gen2).
C) Add a parameter to the pipeline and use it to specify the folder name for each run.
D) Manually rename the destination folder in OneLake after the pipeline completes.

3. You have previously run a pipeline containing multiple activities. What's the best way to check how long each individual activity took to complete?
A) Rerun the pipeline and observe the output, timing each activity.
B) View the run details in the run history.
C) View the Refreshed value for your lakehouse's default dataset.
D) Check the Capacity Metrics app in the Admin portal.

4. If you are already familiar with a specific Azure service, which one shares the same architecture as Microsoft Fabric data pipelines?
A) Azure SQL Database.
B) Azure Data Factory.
C) Azure Functions.
D) Azure Active Directory.

5. What are the two broad categories of activities available in a Fabric pipeline?
A) Ingestion and Visualization.
B) Data transformation and Control flow.
C) Extraction and Storage.
D) Monitoring and Security.

6. Which tool would you use to ingest data from an external source into a lakehouse file or table without applying transformations?
A) Notebook activity.
B) Copy Data activity.
C) Stored procedure activity.
D) KQL Queryset.

7. How can you increase the reusability of your pipelines so they can handle different values for each execution?
A) By duplicating the pipeline for every new value.
B) By using parameters.
C) By hard-coding values into the activities.
D) By creating a new workspace for every unique execution.

8. Which feature provides a guided graphical experience to configure a data source and destination for a copy operation?
A) The Copy Data tool.
B) The Advanced editor.
C) The Spark engine.
D) The API Management layer.

9. What should you use if you need to apply transformations while the data is being ingested from multiple sources?
A) Copy Data activity.
B) Data Flow activity (Dataflow Gen2).
C) Delete data activity.
D) Eventstream.

10. What is the purpose of "Control flow" activities in a pipeline?
A) To define the hardware specifications of the cluster.
B) To implement loops, conditional branching, or manage variables.
C) To create Power BI reports automatically.
D) To regulate the physical temperature of the data center.

11. Where can you find predefined data scenarios to quickly generate a new pipeline?
A) The Monitor hub.
B) The Templates tile.
C) The SQL database settings.
D) The Microsoft 365 Admin Center.

12. Before running a pipeline, which option should you use to check that the configuration is valid?
A) The Schedule option.
B) The Validate option.
C) The Run history option.
D) The "Save As" option.

13. What unique identifier can be used to review specific settings and status for a particular pipeline execution?
A) The Workspace Name.
B) The Run ID.
C) The Tenant ID.
D) The User Profile Name.

Answers

1. B
2. C
3. B
4. B
5. B
6. B
7. B
8. A
9. B
10. B
11. B
12. B
13. B